DREAMGEN: Unlocking Generalization in Robot Learning through Video World Models

Joel Jang, Seonghyeon Ye, Zongyu Lin, Jiannan Xiang, Johan Bjorck, Yu Fang,
Fengyuan Hu, Spencer Huang, Kaushil Kundalia, Lin Yen-Chen, Loic Magne,
Ajay Mandlekar, Avnish Narayan, You Liang Tan, Guanzhi Wang, Jing Wang,
Qi Wang, Yinzhen Xu, Xiaohui Zeng, Kaiyuan Zheng, Ruijie Zheng, Ming-Yu Liu,
Luke Zettlemoyer, Dieter Fox, Jan Kautz, Scott Reed, Yuke Zhu, Linxi Fan

NVIDIA

University of Washington

Citation

- Jang, J., Ye, S., Lin, Z., Xiang, J., Bjorck, J., Fang, Y., Hu, F., Huang, S., Kundalia, K., Yen-Chen, L., Magne, L., Mandlekar, A., Narayan, A., Tan, Y. L., Wang, G., Wang, J., Wang, Q., Xu, Y., Zeng, X., Zheng, K., Zheng, R., Liu, M.-Y., Zettlemoyer, L., Fox, D., Kautz, J., Reed, S., Zhu, Y., & Fan, L. (2025).
- **DREAMGEN: Unlocking Generalization in Robot Learning through Video World Models**.
- arXiv preprint *arXiv:2505.12705* [cs.RO], 17 Jun 2025.
- URL: `https://research.nvidia.com/labs/gear/dreamgen`

Research Area and Background

- Robot foundation models trained on large-scale human teleoperation data for dexterous real-world tasks.
- Current paradigm: imitation learning from manually collected demonstrations for each task and environment.
- Synthetic data in simulation: scalable but suffers from sim2real gap and limited physical realism.
- Recent advances in text-to-video and video world models (e.g., WAN2.1, Cosmos, Hunyuan, CogVideoX).
- Growing interest in learning robot policies from unlabeled videos and latent actions.

Goal of the Research

- Develop a **simple, scalable pipeline** to generate realistic robot training data using video world models.
- Enable **behavior generalization**: learn entirely new verbs and skills from synthetic videos only.
- Enable **environment generalization**: transfer to novel environments using teleoperation data from a single setup.
- Reduce dependence on manual teleoperation while maintaining or improving policy performance.
- Provide a **benchmark** (DreamGen Bench) to evaluate video world models for robotics.

State-of-the-Art and Limitations

- **Simulation-based synthetic data**
  - Efficient but limited by sim2real gap and difficulty modeling liquids, deformables, articulated objects.
  - Often requires heavy manual engineering or TAMP-based systems.
- **Image/video augmentation for robots**
  - Uses in-painting, diffusion, video2video to diversify visuals.
  - Limited diversity in robot motions; mainly improves visual robustness.
- **Video world models as planners**
  - Prior work uses them at test-time for planning or high-level trajectory generation.
  - Not primarily used as large-scale *data generators* for offline policy training.

Research Motivation

- Manual teleoperation for each new task and environment is costly and not scalable.
- Video world models contain strong priors:
    - Physical reasoning and naturalistic motion.
    - Language grounding and scene understanding.
- If adapted to a target robot, they can generate **photorealistic, physically plausible** robot videos.
- These videos lack actions; need a way to recover **pseudo-actions** to make them usable for policy learning.
- Aim: treat video world models as **synthetic data generators** to unlock scalable robot learning.

Problem Definition

- Input:
  - Limited human teleoperation trajectories for a given robot embodiment.
  - A pretrained text-to-video world model.
- Desired output:
  - Visuomotor policies that generalize to:
    - New behaviors (novel verbs).
    - New environments (unseen scenes).
- Constraints:
  - Minimal additional teleoperation data.
  - Use only video world models and pseudo-action labeling for synthetic data.
- Core problem: How to convert video-only generations into **neural trajectories** (video + actions) suitable for policy training.

DREAMGEN Overview

- 4-stage pipeline to create **neural trajectories**:
  1. Fine-tune video world model on robot teleop data.
  2. Roll out model with initial frames and language prompts.
  3. Label pseudo-actions via IDM or latent action model.
  4. Train visuomotor policies on neural trajectories.
- Designed to be general across robots, tasks, and environments.

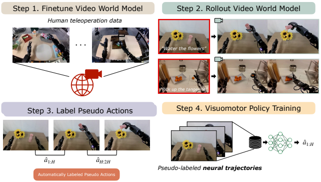


Figure 2: DREAMGEN pipeline: fine-tune, rollout, pseudo-action labeling, policy training.

Generalization Concept

- Figure on page 1 illustrates:
    - Contact-rich data augmentation.
    - New behavior generalization.
    - New environment generalization.
- Only teleoperation data: single pick-and-place task in one environment.
- DREAMGEN enables:
    - 22 new behaviors on GR1 humanoid.
    - Transfer to 10 unseen environments.

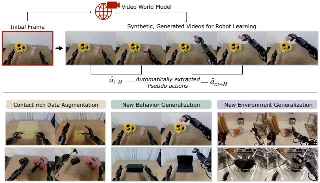


Figure 1: Generalization through DREAMGEN using synthetic videos.

Step 1: Video World Model Fine-tuning

- Base models: primarily WAN2.1; also Hunyuan, CogVideoX, Cosmos for benchmarking.
- Fine-tune on human-teleoperated robot trajectories for target embodiment.
- Use LoRA to preserve internet-scale prior knowledge and avoid catastrophic forgetting.
- For multi-view datasets (RoboCasa, DROID):
  - Concatenate views into a 2×2 grid (left, right, wrist, black).
- Evaluate fine-tuning quality via:
  - Instruction following.
  - Physics following (see DreamGen Bench).

Step 2: Video World Model Rollout

- Generate synthetic robot videos conditioned on:
  - Initial frame o_1.
  - Language instruction i.
- Simulation:
  - Sample new initial frames by randomizing object locations and environments.
- Real world:
  - Manually capture initial frames with randomized object positions.
  - For environment generalization: capture frames from 10 new environments.
- For behavior generalization:
  - Manually design novel behavior prompts (new verbs).
  - Include prompts from DreamGen Bench.

Step 3: Pseudo-Action Labeling

- Two approaches:
  - **Inverse Dynamics Model (IDM)** actions.
  - **Latent actions** via LAPA.
- IDM:
  - Diffusion transformer with SigLIP-2 vision encoder.
  - Conditioned on two frames; predicts action chunks between them.
  - Trained with flow-matching objective on same data as video model.
  - Sliding window over video to produce sequence $\hat{a}_t : \hat{a}_{t+H}$.
- Latent actions (LAPA):
  - Transformer encoder-decoder with VQ-VAE objective.
  - Condition on current and future frame (1s ahead).
  - Use pre-quantized continuous embedding as latent action.

Pseudo-Action Architectures

extracted/figure_3a.png

(a) Inverse Dynamics Model (IDM).

extracted/figure_3b.png

(b) LAPA latent action model.

Step 4: Policy Training on Neural Trajectories

- Neural trajectory: sequence of images and pseudo-actions:

$$(o_t, i_t, \hat{a}_t : \hat{a}_{t+H})$$

- Policies condition on:
  - Image observation o_t.
  - Language instruction i_t.
  - State inputs are set to zero (no state in neural trajectories).
- Target: predict pseudo-actions $\hat{a}_{t:t+H}$ (IDM or latent).
- Policy architectures:
  - Diffusion Policy.
  - π_0.
  - GR00T N1.
- Training regimes:
  - Co-training with real trajectories (1:1 sampling).
  - Training solely on neural trajectories (IDM actions).

Experimental Design

- **Three main applications**:
  - Training data augmentation for existing tasks.
  - Behavior generalization to novel verbs.
  - Environment generalization to novel scenes.
- **Embodiments**:
  - Fourier GR1 humanoid.
  - Franka Emika Panda.
  - SO-100 low-cost arm.
- **Policy backbones**: Diffusion Policy, π_0, GR00T N1.
- **Evaluation**: success rate with task-specific partial credit schemes, 10 rollouts per checkpoint.

Datasets Used

- **RoboCasa** (simulation)
  - 24 kitchen manipulation tasks with multi-view observations.
  - 1,200 human demonstrations for video model training.
- **GR1 teleoperation**
  - 2,884 pick-and-place trajectories in a single lab environment.
  - Additional 100 trajectories per dexterous task (hammering, wiping, folding, stacking).
- **DROID** (Franka)
  - 49,895 trajectories for video model and IDM training.
  - 100 teleop trajectories per real-world Franka task.
- **SO-100** (LeRobot)
  - Trimmed and split teleop videos into individual trajectories.
  - 10 and 13 videos for two tasks, expanded into 68 and 44 trajectories.

Scaling Neural Trajectories in RoboCasa

- Vary number of neural trajectories up to 240k (333× original demos).
- Three ground-truth data regimes:
  - Low (720), mid (2.4k), high (7.2k) trajectories.
- Co-training with neural trajectories improves success across all regimes.
- Log-linear relationship between #neural trajectories and performance.
- Training only on neural trajectories (IDM) achieves 20.6% average success (24 tasks).

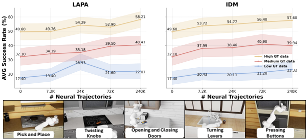


Figure 4: RoboCasa success vs. number of neural trajectories (LAPA and IDM).

Real-World Data Augmentation Results

- GR1 humanoid: 4 tasks (hammering, wiping, folding, stacking).
- Franka: 3 tasks (pick-and-place, cube stacking, tool usage).
- SO-100: 2 tasks (strawberry picking, tic-tac-toe).
- Low-data regime:
  - 10 trajectories per task (25 for GR1 folding).
  - 10 and 13 trajectories for SO-100 tasks.
- Co-training with neural trajectories (1:1) consistently improves success for all policies and robots.

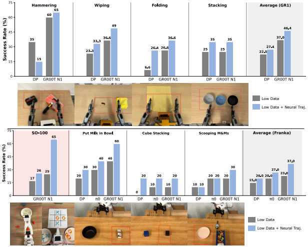


Figure 5: Real-world evaluation with and without neural trajectories.

Behavior and Environment Generalization Setup

- Train video world model on:
  - 2,884 GR1 pick-and-place trajectories in a single environment.
- Behavior generalization:
  - 14 novel behavior tasks in seen environment (e.g., pouring, opening doors, using tools).
  - 50 neural trajectories per task; no additional teleoperation.
- Environment generalization:
  - 10 novel environments with new initial frames.
  - Both seen behaviors (pick-and-place variants) and novel behaviors.
- Base policy: GR00T N1 trained on pick-and-place only vs. GR00T N1 with DREAMGEN.

Generalization Results: Behaviors and Environments

- **Seen environments, novel behaviors** (14 tasks):
  - GR00T N1 baseline: 11.2% average success.
  - With DREAMGEN: 43.2% average success.
- **Novel environments, seen behaviors** (6 tasks):
  - Baseline: 0.0% average success.
  - With DREAMGEN: non-trivial success (e.g., 30–45% on several tasks).
- **Novel environments, novel behaviors** (7 tasks):
  - Baseline: 0.0% average success.
  - With DREAMGEN: 28.5% average success.
- Demonstrates zero-to-one improvements from pick-and-place-only teleoperation data.

Table: Generalization Success Rates

Model	Seen Environments, Novel Behaviors														
Model	Open Microwave	Open Macbook	Close Lunchbox	Hit Tambourine	Hit Keyboard	Grab button	Pour Water	Water flowers	Light Candle	Use Vacuum	Iron shirt	Take Spoon Out	Unroll mat	Move Mouse	Average
GR00T N1	0	0	0	5	0	45	40	50	10	0	0	7	0	0	11.2
w/ DreamGen	**23**	**45**	**10**	**15**	**90**	**75**	**55**	**95**	**15**	**55**	**20**	**17**	**55**	**35**	**43.2**
Examples															

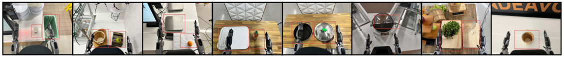
Model	Novel Environments, Seen Behaviors						Novel Environments, Novel Behaviors							
Model	Pick up Tangerine	Box sandwich	Weigh the Orange	Put cup in trash	Put pear in basket	Put sauce on tray	Water Flowers	Lift Basket	Swirl Around Spoon	Use Whisk	Close soup container	Uncover Pot	Cover Pot	Average
GR00T N1	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0
w/ DreamGen	**30**	**10**	**20**	**45**	**35**	**45**	**15**	**55**	**15**	**25**	**55**	**30**	**35**	**28.5**
Examples														

Table 1: Success rates across new behaviors and environments for GR1 humanoid.

DreamGen Bench: Benchmark Design

- Goal: evaluate video world models for robotics along:
  - **Instruction Following** (IF).
  - **Physics Alignment** (PA).
- Datasets:
  - RoboCasa (Franka, simulation).
  - GR1 humanoid (real).
- Metrics:
  - IF: binary scores from Qwen2.5-VL (and GPT-4o) on instruction adherence.
  - PA: scores from VideoCon-Physics and Qwen2.5-VL on physical plausibility.
- Human evaluation used to validate automatic metrics (high Pearson correlation > 0.9).

DreamGen Bench: Model Comparison

- Evaluated models:
  - Hunyuan, CogVideoX, WAN2.1, Cosmos.
  - Zero-shot and fine-tuned variants.
- Fine-tuned models substantially outperform zero-shot on both IF and PA.
- Cosmos-sft and WAN2.1-sft achieve highest combined scores across datasets.
- Benchmark provides a low-cost proxy for downstream policy performance.

Table: DreamGen Bench Statistics and Results

Dataset Statistics

Dataset	RoboCasa	GR1		
Train (# trajs)	1200	100		
Eval (# frames)	48	Object: 50	Behavior: 47	Env: 30

Results

	IF			PA	IF			PA	IF			PA	IF			PA
	GPT	Qwen	Hu		GPT	Qwen	Hu		GPT	Qwen	Hu		GPT	Qwen	Hu	
Hunyuan-zero	1.0	0.0	-	0.0	0.0	0.0	-	0.0	0.0	2.1	-	2.1	0.0	0.0	-	0.0
CogVideoX-zero	0.0	0.0	-	0.0	0.0	0.0	-	0.0	0.0	0.0	-	0.0	0.0	0.0	-	0.0
WAN2.1-zero	0.0	0.0	-	0.0	0.0	2.0	-	2.0	0.0	2.1	-	2.1	0.0	6.7	-	6.7
Cosmos-zero	**4.2**	**22.9**	-	**22.9**	0.0	**32.0**	-	**32.0**	**6.4**	**31.9**	-	**31.9**	**3.5**	**24.1**	-	**24.1**
Hunyuan-sft	68.8	8.3	81.3	44.8	38.0	26.0	52.0	39.0	38.3	10.6	14.9	12.8	27.6	27.6	43.2	35.4
CogVideoX-sft	72.9	10.4	79.2	44.8	72.0	38.0	72.0	55.0	44.0	28.0	21.3	24.7	55.2	41.4	61.1	51.3
WAN2.1-sft	77.1	18.8	91.7	55.3	72.0	58.0	80.0	69.0	**72.3**	55.3	**74.5**	**64.9**	48.3	**65.5**	**67.4**	**66.5**
Cosmos-sft	**79.2**	**29.2**	**93.8**	**61.5**	**90.0**	**62.0**	**84.0**	**73.0**	59.6	**61.7**	68.1	**64.9**	**69.0**	**65.5**	53.3	59.4

Table 2: Instruction following and physics alignment scores for different video models.

Correlation with Downstream Performance

- Train RoboCasa policies using only neural trajectories from each video model.
- Compute DreamGen Bench score as average of IF (GPT) and PA.
- Plot RoboCasa success vs. DreamGen Bench score.
- Observe positive correlation:
  - Better world models (higher IF/PA) yield better robot policies.
- Supports using DreamGen Bench as a proxy for model selection.

extracted/figure_6.png

Figure 6: Correlation between DreamGen Bench and RoboCasa performance.

Conclusion

- DREAMGEN: a 4-stage pipeline turning video world models into scalable synthetic data generators.
- Neural trajectories (videos + pseudo-actions) enable:
  - Strong data augmentation in simulation and real world.
  - Behavior generalization to 22 new skills on GR1.
  - Environment generalization from a single teleop environment to 10 new ones.
- DreamGen Bench links video model quality (IF, PA) to downstream robot performance.
- Establishes a promising direction for scaling robot learning beyond manual data collection.

Limitations and Future Work

- Tasks are relatively simple; do not cover full dexterous capabilities of robots.
- Significant compute cost:
  - 240k RoboCasa samples took 54 hours on 1500 NVIDIA L40 GPUs.
- Requires manual selection of initial frames; automation is future work.
- Automatic evaluators (VideoCon-Physics, VLMs) can hallucinate on physical realism.
- Future directions:
  - More diverse behaviors and video-language pairings.
  - Richer control for complex dexterous tasks.
  - More efficient video world models and evaluation methods.

Thank You!

Questions?